UWE-4 Operational Strategy

Target Selection & Passes

Phase 1: The Selection Funnel

Selection Criteria

To build a reliable detector, we need reliable data. We filter the entire amateur fleet through five critical layers to identify our "Golden Targets."

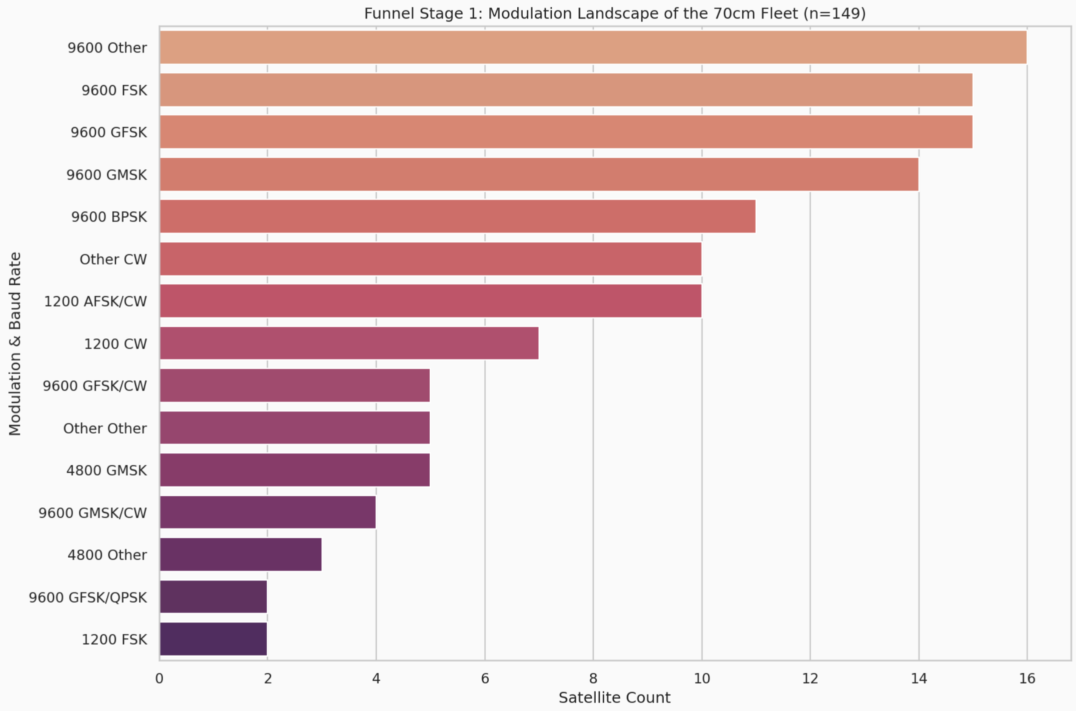
- **Status:** Must be confirmed 'Alive' in SatNOGS DB.
- **Band:** 433-438 MHz (70cm Amateur Band) for antenna compatibility.
- **Modulation:** High-rate 9600 bps FSK/GFSK for high-fidelity signal profiles.
- **Support:** Explicitly supported by `satnogs-decoders` (Kaitai) for automated parsing.
- **Visibility:** High-elevation passes (>30°) for clean, noise-free reception.

Stage 1: The Fleet Landscape

We analyzed 149 active satellites in the target band to identify dominant communication standards.

Observations:

- Huge variety in modulation types.
- Many legacy satellites use low-rate AFSK or CW.
- No single "universal" standard exists across the entire fleet.



Stage 2: Technical Filtering

We narrow the funnel by applying technical constraints required for edge anomaly detection.

Funnel Step	Constraint	Remaining
1. Bandwidth	70cm Amateur (433-438 MHz)	149
2. Decodeable	Explicitly supported by `satnogs-decoders`	7
3. High Rate	9600 bps GMSK/GFSK/FSK	2

Result: We converged on a “Golden Cohort” of 2 targets that provide high-fidelity telemetry for ML training.

Phase 2: Operational Planning

Heuristic: "Total Observable Time"

Pass count isn't enough. We rank satellites by the **total duration** they spend above 30° elevation over 48 hours.

Satellite Scoring

Goal: Maximize data collection opportunity

Logic: Sum(Duration of all passes $> 30°$)

Result: Top Operational Targets

Stage 3: Operational Convergence

Ranked by "Total Observable Time" over Beni Suef (>30° Elevation).

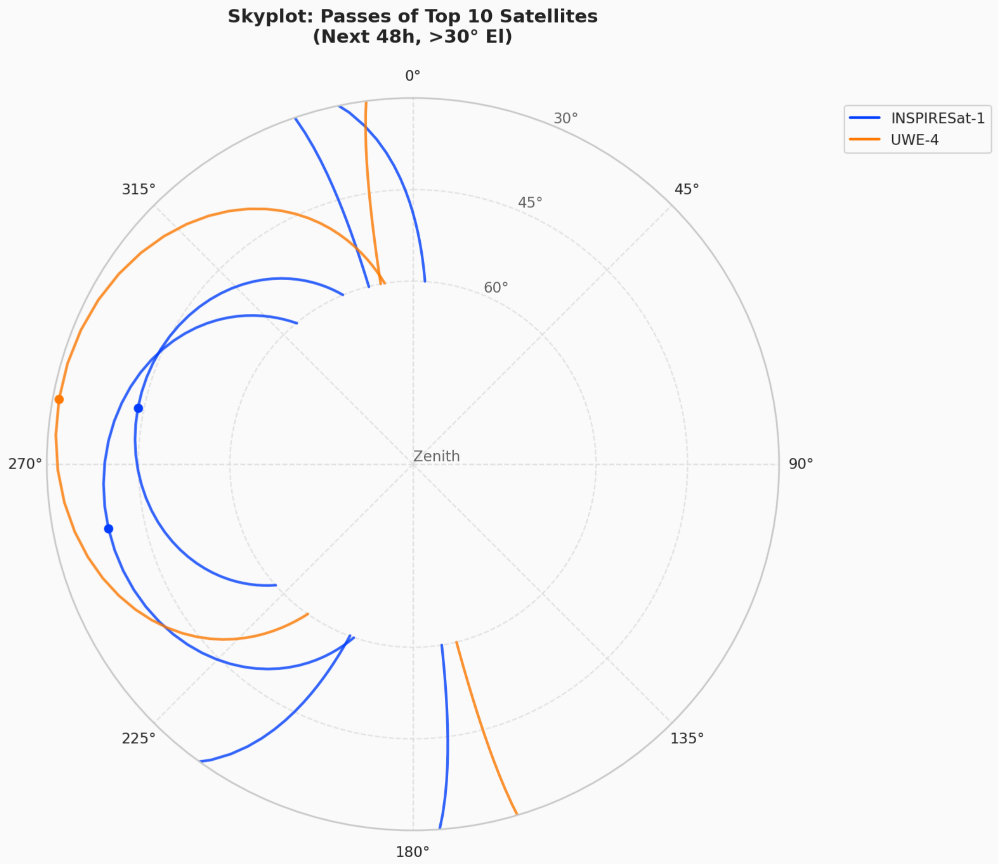
Satellite	Passes (48h)	Total Mins	Max El
INSPIRESat-1	4	7.8 m	85.8°
UWE-4	2	5.8 m	87.4°

Why UWE-4? Despite having fewer passes than INSPIRESat-1, UWE-4 is our primary "Golden Path" due to its rich, multi-sensor telemetry (Panel temps, Battery currents, OBC health) which is essential for complex anomaly detection.

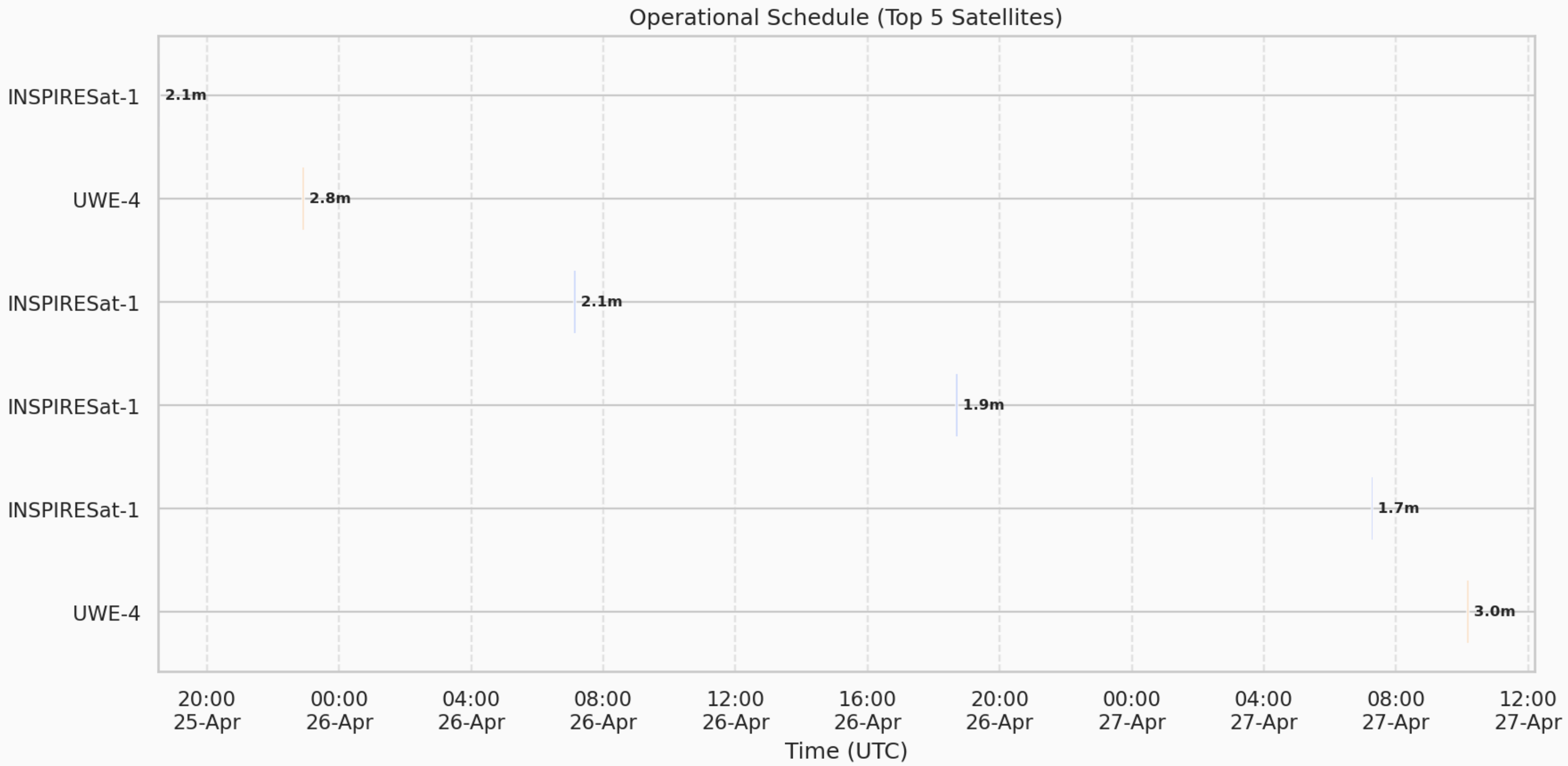
***Data Engineering Reality Check:** BugSat-1 was dropped despite high visibility due to undocumented protocol variations (`US37` payload header) causing Kaitai parser failures. ML requires clean data; UWE-4 provides perfect compatibility and rich thermal/power telemetry.

Operational Reality: Skyplot

Where to point the antenna for the Top targets.



When to operate the ground station (Next 48 Hours).



The V-Model Strategy

Two distinct environments sharing a single **Shared Core**.

A. The Lab (Offline)

- Establish Baseline from history
- Source: SatNOGS DB
- **Action:** Train Autoencoder

B. The Watchdog (Edge Deployment)

- Live anomaly detection (5 Docker Microservices)
- Source: Live Antenna -> SDR -> MQTT
- **Action:** FastAPI backend inference & SvelteKit dashboard

Standardization: The Golden Features

A universal interface for heterogeneous satellite hardware.

Feature	**Unit**	**Description**
`batt_voltage`	Volts	Standardized from mV/ADC
`batt_current`	Amps	Charge/Discharge rate
`power_consumption`	Watts	Spacecraft power draw
`temp_obc`	Celsius	Main computer temp
`temp_batt_a/b`	Celsius	Battery thermal health
`temp_panel_z`	Celsius	Orbit-phase context
`uptime`	Seconds	Time since reset